



CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 15: Similarity (With Applications to Maps and Models)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Similar triangles

1. $\triangle ABC \sim \triangle EFG$ with $\angle ABC = \angle EFG = 60^\circ$, $AB = 15$ cm, $BC = 3$ cm and $EF = 75$ cm. Find FG .
(a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm **ICSE 2025, 1 mark**

SOLUTION

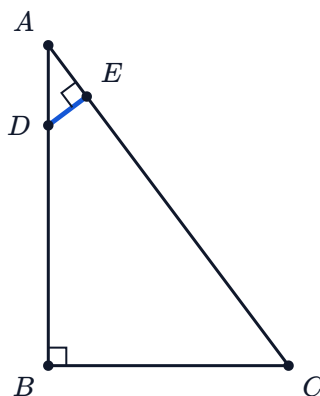
Corresponding sides are proportional: $\frac{AB}{EF} = \frac{BC}{FG}$.

$$\frac{15}{75} = \frac{3}{FG}, \text{ so } FG = 15 \text{ cm.}$$

Answer: (a)

2 Maps and models

2. In $\triangle ABC$, $\angle ABC = 90^\circ$, $AB = 20$ cm and $AC = 25$ cm. D is a point on AB and E is a point on AC with $DE \perp AC$ and $DE = 3$ cm. (a) Prove that $\triangle ABC \sim \triangle AED$. (b) Find BC , AD and AE . (c) $BCED$ is a plot of land drawn on a map, and its actual area is 576 m^2 . Find the scale factor of the map. **ICSE 2025, 4 marks**



SOLUTION

(a) In $\triangle ABC$ and $\triangle AED$: $\angle A$ is common and $\angle ABC = \angle AED = 90^\circ$. So $\triangle ABC \sim \triangle AED$ (AA).

(b) $BC = \sqrt{25^2 - 20^2} = 15$ cm. From the similar triangles, $\frac{AB}{AE} = \frac{BC}{ED} = \frac{AC}{AD} = \frac{15}{3} = 5$.

$AE = \frac{20}{5} = 4$ cm and $AD = \frac{25}{5} = 5$ cm.

(c) Area of $BCED$ on the map = area of $\triangle ABC$ – area of $\triangle AED = \frac{1}{2} \times 20 \times 15 - \frac{1}{2} \times 4 \times 3 = 150 - 6 = 144 \text{ cm}^2$.

Actual area = $576 \text{ m}^2 = 5760000 \text{ cm}^2$. $k^2 = \frac{144}{5760000} = \frac{1}{40000}$, so $k = \frac{1}{200}$.

Answer: (b) $BC = 15$ cm, $AD = 5$ cm, $AE = 4$ cm (c) scale factor $1 : 200$

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